R13

Code No: 115AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech III Year I Semester Examinations, May/June - 2019
PRINCIPLES OF PROGRAMMING LANGUAGES
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

used paper 21-06-2019 PM

PART - A

(25 Marks)

1.a)	Explain about the Virtual Machine.	[2]
b)	What are the uses of attribute grammar?	[3]
c)	Explain about the problems in unconditional branching.	[2]
d)	Explain about the enumerated data type.	[3]
e)	What are the characteristics of subprograms?	[2]
f)	Explain about coroutines.	[3]
g)	Why does Java not have a destructors?	[2]
h)	What are the applications of logic programming?	[3]
i)	Describe the scoping rule in ML.	[2]
j)	Explain about the fundamentals of FPL.	[3]

PART - B

(50 Marks)

2.a)	Explain about the preconditions and postconditions of a given statement mean in axiomatic semantics.	
b)	Describe the steps involved in the language evaluation criteria.	[5+5]
	OR	
3.a)	Explain the different categories of languages.	
b)	Draw and explain the flow chart for compilation process.	[5+5]
4.a)	Explain about the mixed-mode assignments that are used in Ada and Java Languages.	
b)	Explain about the type compatibility with an example.	[5+5]
	OR	
5.a)	What is type checking? Differntiate between static and dynamic type checking and give their relative advantages.	
b)	Define an array? Explain how to initialize an array? Explain the different types of arrays.	[5+5]

6.a) Describe about the static and dynamic scope of variables with an example.
b) Define sub program. What are the distinct categories of subprograms. [5+5]

OR

7.a) Explain about the generic subprograms in Ada with an example.
b) Explain about the semantic models of parameter passing. [5+5]

8.a) Explain about the concurrency in Ada 95.
b) Explain the basic elements of prolog. [5+5]

OR

9.a) Explain how to handle the exceptions in Ada.
b) What are the design issues of abstract data types. [5+5]

10.a) Explain about the internal representation of two LISP lists.
b) Describe the scoping rule in common LISP and Haskell. [5+5]

OR

11.a) Compare the functional programming languages with imperative languages.
b) Write a LISP function Fib(n) that computes nth Fibonacci number. [5+5]

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